

# 6

## Complex Numbers

### 6.1 Basic Operations on Complex Numbers

#### Why complex numbers?

Here is a historical development of numbers leading naturally to need for complex number.

- ① In  $\mathbb{N}$  we cannot solve  $x + 2 = 1$ .
- ② In  $\mathbb{Z}$  we cannot solve  $2x = 1$ .
- ③ In  $\mathbb{Q}$  we cannot solve  $x^2 = 2$ .
- ④ In  $\mathbb{R}$  we cannot solve  $x^2 + 1 = 0$ .

Complex numbers were coined to define numbers constituting solutions of polynomial equations like  $x^2 + 1 = 0$ . More generally,

#### The quadratic formula

Consider the quadratic equation  $ax^2 + bx + c = 0$ . Recall that the two roots of the quadratic equation  $ax^2 + bx + c = 0$  are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (6.1)$$

Equation (6.1) above is called the **quadratic formula**. The quantity  $\Delta = b^2 - 4ac$  that appears under the square root is called the **discriminant** of the quadratic equation or polynomial. The nature of the roots of the quadratic equation depends on the sign of the discriminant.

- ① If  $b^2 - 4ac > 0$  then these two roots are real and distinct; graphically they are at the two points where the curve  $y = ax^2 + bx + c$  cuts the  $x$ -axis (or the  $x$ -intercepts of curve  $y = ax^2 + bx + c$ ).
- ② If  $b^2 - 4ac = 0$  then we have one real root and the curve just touches the  $x$ -axis here.

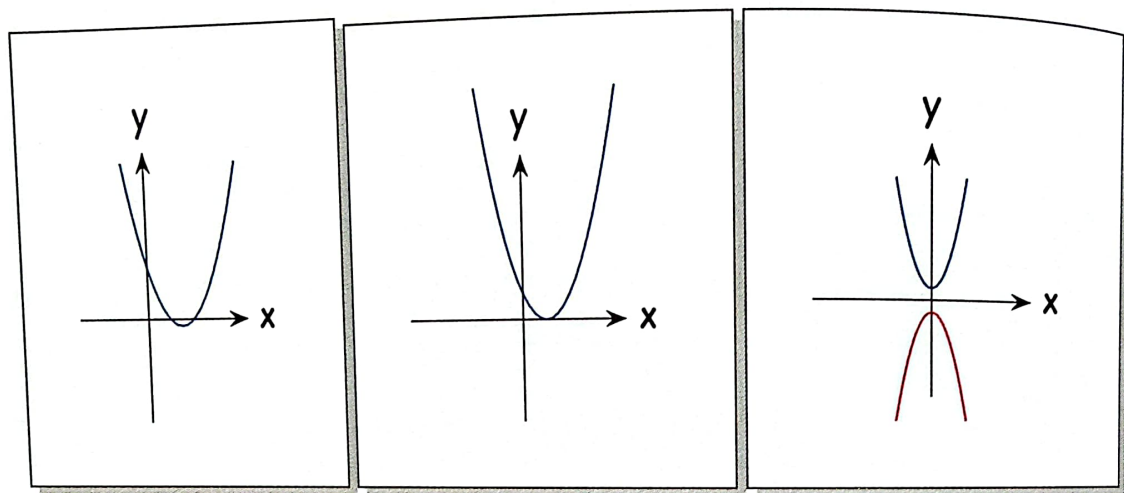


Figure 6.1:  $b^2 - 4ac > 0$ ,  $b^2 - 4ac = 0$  and  $b^2 - 4ac < 0$  respectively

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But what happens when  $b^2 - 4ac < 0$ ? From the graphical point of view the curve  $y = ax^2 + bx + c$  lies entirely above or below the  $x$ -axis.

The upshot is that there are no real solutions to the equation since no real squares can give the negative  $b^2 - 4ac$ . But nothing stops us from writing the two roots of the quadratic equation in (6.2) in the form:

$$x = \alpha \pm \beta\sqrt{-1} \quad (6.2)$$

where  $\alpha = -\frac{b}{2a}$  and  $\beta = \frac{\sqrt{4ac - b^2}}{2a}$  are real numbers.

We conventionally write  $i$  for  $\sqrt{-1}$  as first introduced by the Swiss mathematician Leonhard Euler (1707-1783). The symbol  $i$  is called the **imaginary unit** and defined by the property  $i^2 = -1$ . Sometimes the symbol  $j$  instead of  $i$  is used—for example in circuit theory where  $i$  is reserved for a current.

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$x = \sqrt{-1} = \pm i$ . The solutions of the equation  $x^2 + 1 = 0$  is

Multiples of  $i$  are called **pure imaginary numbers**. A general complex number is the sum of a multiple of  $1$  and a multiple of  $i$  such as  $z = 2 + 3i$ . We often use the notation  $z = a + bi$ , where  $a$  and  $b$  are real.

More formally,

DEFINITION 6.1

**Complex number** A complex number is any number of the form  $z = a + ib$ , where  $a$  and  $b$  are real numbers.

The notations  $a + ib$  and  $a + bi$  are used interchangeably. The real number  $a$  in  $z = a + ib$  is called the **real part** of  $z$  while  $b$  is called the **imaginary part** of  $z$  and we use operators for extracting  $a, b$  from  $z$  to write  $a = \text{Re}(z)$  and  $b = \text{Im}(z)$ .

DEFINITION 6.2

**Equality** Two complex numbers  $z_1 = a_1 + ib_1$ ,  $z_2 = a_2 + ib_2$  are equal if  $a_1 = a_2$  and  $b_1 = b_2$ .

For instance, since  $\log_{10} 0.1 = -1$  and  $0.5 = \frac{1}{2}$ . Then the complex numbers

$$-1 + 0.5i = \log_{10} 0.1 + \frac{1}{2}i$$

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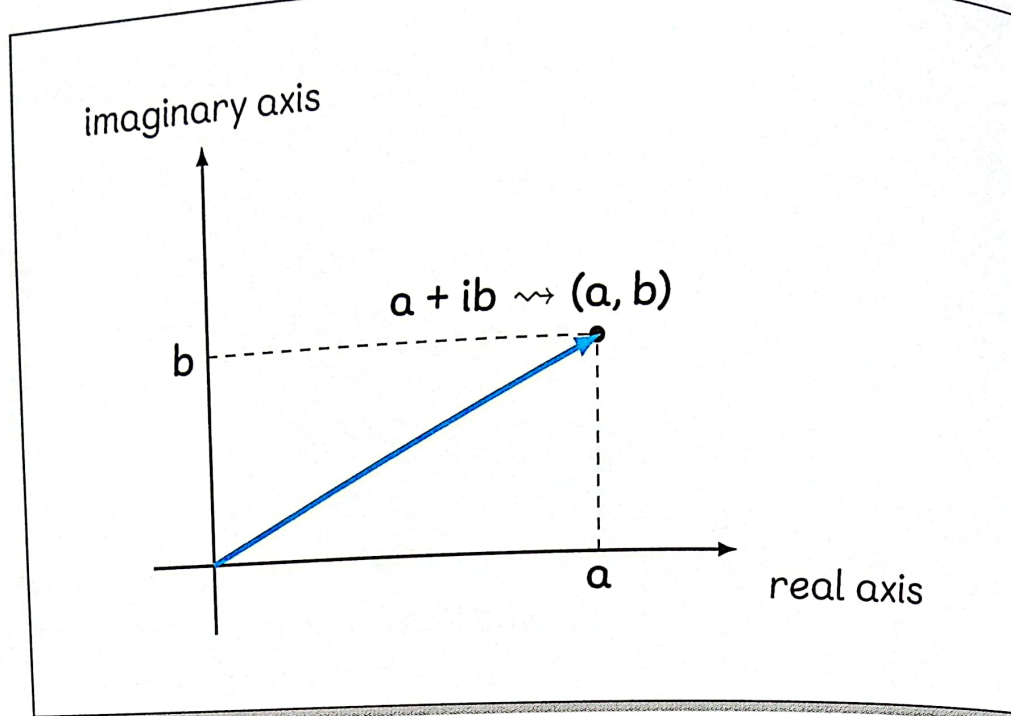
We denote by  $\mathbb{C}$  **the set of all complex numbers**. Note that real numbers are complex numbers: a real number is simply a complex number with no imaginary part.

In other words any real number  $a$  can be written as  $a + i0$ , we see that the set  $\mathbb{R}$  is a subset of  $\mathbb{C}$ .

6.1.1

### The Argand diagram—The complex plane

A complex number  $z = a + ib$  is uniquely determined by a point  $(a, b)$  in the Cartesian coordinate plane. Indeed  $\mathbb{C}$  is sometimes referred to as the **complex plane**, or more commonly we call it an **Argand diagram** (after the Swiss mathematician Jean-Robert Argand).



### EXAMPLE 6.1

The complex number  $3 - 4i$  correspond to the ordered pair  $(3, -4)$ .

Since an ordered pair  $(a, b)$  also corresponds to such a position vector. This leads to the definition of the distance of a complex number  $z = a + bi$  from the origin.

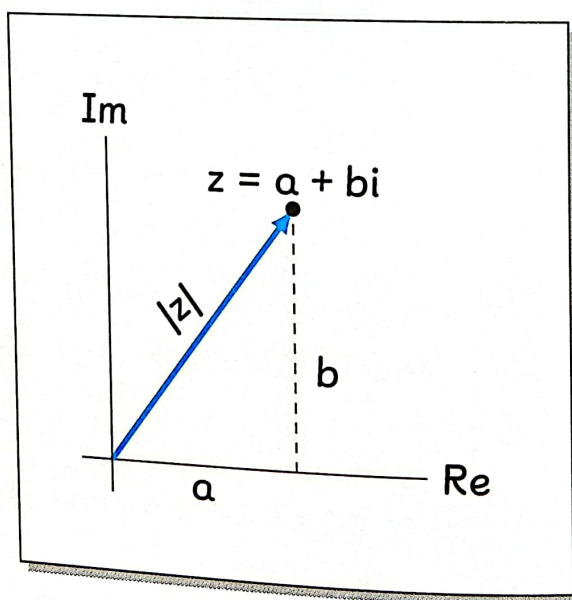


Figure 6.2: Modulus of a complex number

### DEFINITION 6.3

**Modulus** The modulus (length) of a complex number  $z = a + bi$  is the real number

$$|z| = \sqrt{a^2 + b^2}.$$

(6.3)

The modulus of the complex number  $z = 2 - 3i$  from the origin is

$$|z| = \sqrt{2^2 + (-3)^2} = \sqrt{4 + 9} = \sqrt{13}$$

while the modulus of the complex number  $z = -9i$ ,

$$|z| = |-9i| = \sqrt{(-9)^2} = 9.$$

## Polar Form of Complex Numbers

Recall in polar coordinates the point  $(a, b)$  is described by the ordered pair  $(r, \theta)$  values.

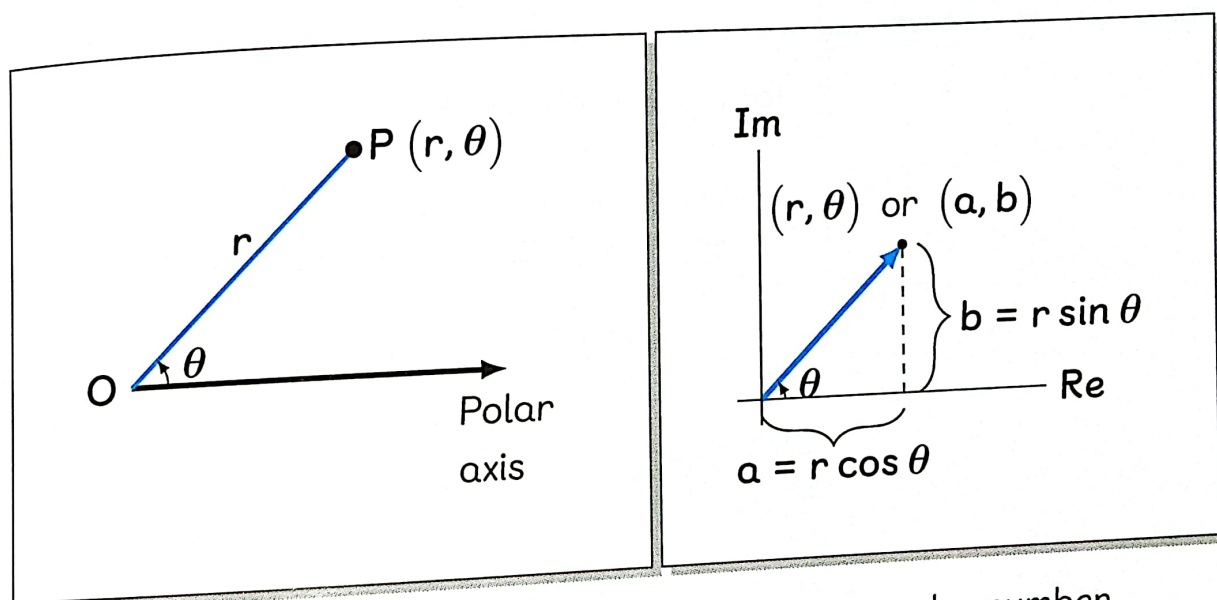


Figure 6.3: The polar representation of a complex number

Suppose the polar coordinate system is superimposed on to the complex plane with the polar axis coinciding with the positive real axis and the pole  $O$  at the origin. It is clear from Fig 6.3 that the variables  $x, y, r$  and  $\theta$  are related by the equations

$$r = |z| = \sqrt{a^2 + b^2}, \quad a = |z| \cos \theta = r \cos \theta, \quad b = |z| \sin \theta = r \sin \theta.$$

These equations enable us to express a nonzero complex number using polar coordinates as follows

$$z = a + ib = r \cos \theta + ir \sin \theta = r (\cos \theta + i \sin \theta). \quad (6.4)$$

We refer to Equation (6.4) as the polar form (or polar representation) of the complex number  $z$ .

**DEFINITION 6.4**

**Argument** The angle  $\theta$  is called the **argument** of  $z$  and is denoted by  $\theta = \arg(z)$ . An argument of a complex number  $z$  must satisfy the equations

$$\cos \theta = \frac{x}{r} \quad \text{and} \quad \sin \theta = \frac{y}{r}.$$

Since sine and cosine are  $2\pi$ -periodic, it follows that the argument of a complex number  $z$  is not unique. Therefore, if  $\theta$  is an argument of  $z$ , then  $\theta = \theta + 2n\pi$ ,  $n = 0, \pm 1, \pm 2, \dots$  are also arguments of  $z$ .

However if we restrict  $\theta$  to the interval  $(-\pi, \pi]$  then the argument is unique.

**DEFINITION 6.5**

**Principal Argument** The principal argument of  $z$  denoted by  $\text{Arg}(z) = \arg(z)$  such that

$$-\pi < \arg(z) \leq \pi$$

for any complex number  $z$ .

Thus in general,  $\arg(z)$  and  $\text{Arg}(z)$  are related as follows

$$\arg(z) = \text{Arg}(z) + 2n\pi, \quad n = 0, \pm 1, \pm 2, \dots$$

**STRATEGY**

In practice we use the relation  $\tan \theta = y/x$  to compute  $\theta$  and since tangent is  $\pi$ -periodic the actual computation values obtained satisfies

$$-\frac{\pi}{2} < \tan^{-1}\left(\frac{y}{x}\right) < \frac{\pi}{2}.$$

Thus, care must be taken to factor in the quadrant in the complex plane where the corresponding complex number is located.

We use the principal argument of the complex number in writing numbers in converting numbers to polar form. That is given a complex number  $z$  the default polar representation

$$z = r(\cos \theta + i \sin \theta)$$

where  $\theta = \text{Arg}(z)$ .

Express  $-\sqrt{3} - i$  in polar form.

**SOLUTION:**

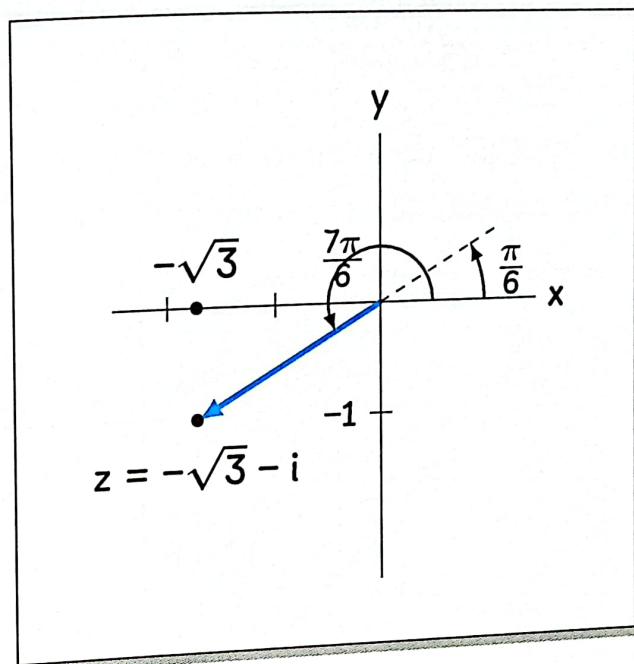


Figure 6.4

Here  $x = -\sqrt{3}$ ,  $y = -1$ . Therefore

$$r = |z| = \sqrt{x^2 + y^2} = \sqrt{-\sqrt{3}^2 + (-1)^2} = \sqrt{3 + 1} = 2.$$

Now  $\frac{y}{x} = \frac{-1}{-\sqrt{3}} = \frac{1}{\sqrt{3}}$  and we obtain  $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$  which is an angle in the first quadrant. However since  $-\sqrt{3} - i$  lies in the third quadrant, we take the solution of  $\tan \theta = \frac{1}{\sqrt{3}}$  to be  $\theta = \arg(z) = \frac{\pi}{6} + \pi = \frac{7\pi}{6}$ .

Thus the polar form of  $z = -\sqrt{3} - i$  is

$$z = 2 \left( \cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} \right).$$

The complex number obtained by changing the sign of  $i$  is a new complex number denoted by  $\bar{z}$ . Formally,

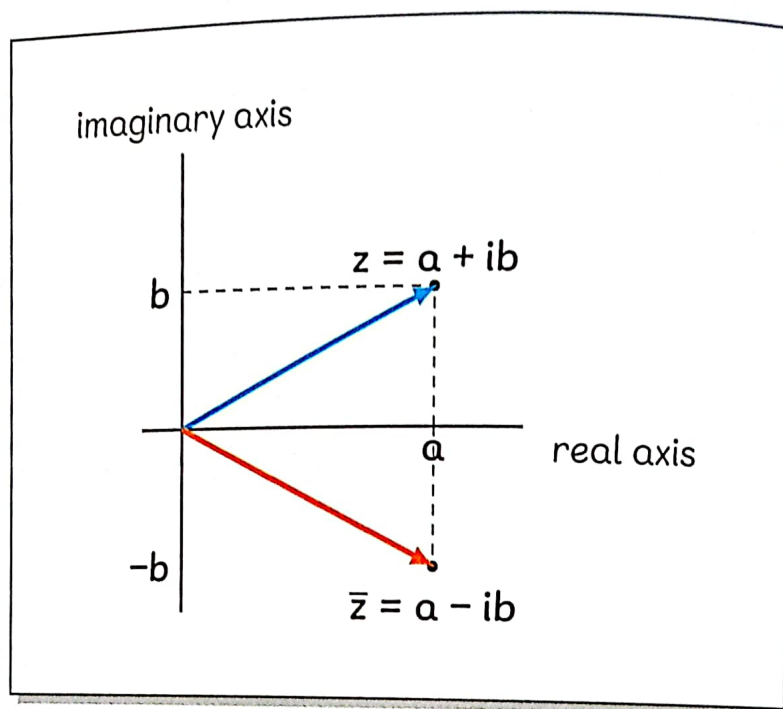
**DEFINITION 6.6**

**Complex conjugate**

Let  $z = a + ib$ . The conjugate of  $z$  is the number  $\bar{z} = a - ib$ .



Geometrically, taking a conjugate is equivalent to reflecting the complex number through the real axis. It follows that real numbers are self-conjugate. That is the complex conjugate of a real number (viewed as complex number) is itself.



## 6.1.2 Arithmetic Operations

We add, subtract, multiply and divide complex numbers much as we would expect.

### Addition and subtraction

Complex numbers addition and subtraction follow the same rules as for ordinary numbers except that the real and imaginary parts are treated separately:

#### DEFINITION 6.7

**Addition and Subtraction** Let  $z_1 = a + bi$  and  $z_2 = c + di$  be complex numbers. Complex numbers addition and subtraction is given by

$$z_1 + z_2 = (a + bi) + (c + di) = (a + c) + (b + d)i$$

and

$$z_1 - z_2 = (a + bi) - (c + di) = (a - c) + (b - d)i.$$

If  $z_1 = 2 + 4i$  and  $z_2 = -3 + 8i$ . Then  $z_1 + z_2$  and  $z_1 - z_2$  can be calculated

as follows:

$$\begin{aligned} z_1 + z_2 &= (2 + 4i) + (-3 + 8i) \\ &= (2 + (-3)) + (4 + 8)i = -1 + 12i. \end{aligned}$$

$$\begin{aligned} z_1 - z_2 &= (2 + 4i) - (-3 + 8i) \\ &= (2 - (-3)) + (4 - 8)i = 5 - 4i. \end{aligned}$$

Since the complex numbers are vectors on the Argand diagrams, addition and subtraction can be represented graphically as in Figure 6.5.

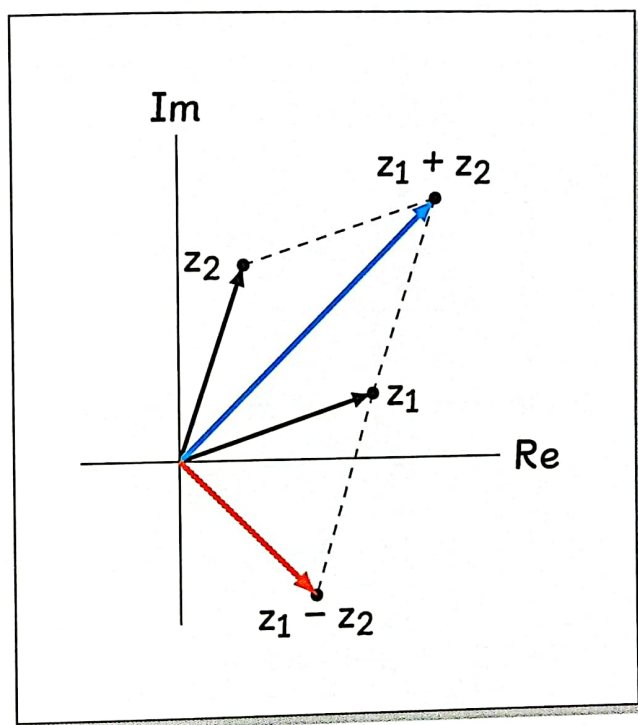


Figure 6.5: addition and subtraction on the Argand diagram

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We have some interesting properties when complex numbers are added or subtracted to their conjugates.

For a complex number  $z = a + ib$

$$\bar{z} + z = (a + ib) + (a - ib) = 2a = 2\operatorname{Re}(z),$$

from which we observe that

$$\operatorname{Re}(z) = \frac{\bar{z} + z}{2}.$$

Similarly we have

$$\bar{z} - z = (a + ib) - (a - ib) = 2ib = 2i\operatorname{Im}(z),$$

so that

$$\operatorname{Im}(z) = \frac{\bar{z} + z}{2i}.$$

## Multiplication and Division

We can multiply complex numbers by expanding the brackets in the usual fashion and using  $i^2 = -1$ ,

### DEFINITION 6.8

**Multiplication** Let  $z_1 = a + bi$  and  $z_2 = c + di$  be complex numbers. Multiplication is defined as follows

$$\begin{aligned} z_1 \cdot z_2 &= (a + bi)(c + di) \\ &= ac + adi + cbi + i^2 bd \\ &= (ac - bd) + (ad + cb)i \end{aligned}$$

One of the products of interest is that of a complex number and its conjugate. That is

$$\begin{aligned} z \cdot \bar{z} &= (a + ib)(a - ib) \\ &= a^2 - iab + iab - i^2 b^2 \\ &= a^2 + b^2. \end{aligned}$$



The product of complex number and its conjugate is always a real number. This is a very useful property in working with quotients of complex numbers.

### DEFINITION 6.9

**Division** Division of two complex numbers  $z_1$  and  $z_2$  is denoted by  $\frac{z_1}{z_2}$  defined by

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{a + ib}{c + id} = \frac{a + ib}{c + id} \cdot \frac{c - id}{c - id} \\ &= \frac{(ac + bd) + i(-ad + cb)}{c^2 + d^2} \\ &= \frac{ac + bd}{c^2 + d^2} + i \frac{cb - ad}{c^2 + d^2}. \end{aligned}$$

Notice we multiply both the numerator and the denominator by the conjugate of the denominator in order to write the result in the form  $a + bi$ .

If  $z_1 = 2 + 4i$  and  $z_2 = -3 + 8i$ , find

(a)  $z_1 \cdot z_2$ .

(b)  $\frac{z_1}{z_2}$ .

**SOLUTION:**

(a)

$$\begin{aligned} z_1 \cdot z_2 &= (2 + 4i)(-3 + 8i) \\ &= (2 \cdot -3 - 4 \cdot 8) + i(2 \cdot 8 + 4 \cdot -3) \\ &= -38 + 4i. \end{aligned}$$

(b)

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{2 + 4i}{-3 + 8i} \\ &= \frac{2 + 4i}{-3 + 8i} \cdot \frac{-3 - 8i}{-3 - 8i} \\ &= \frac{(2 \cdot -3 - 4 \cdot -8) + i(2 \cdot -8 + 4 \cdot -3)}{(-3)^2 + (-8)^2} \\ &= \dots \\ &= \frac{26}{73} - \frac{28}{73}i. \end{aligned}$$

Every nonzero complex number  $z$  has a reciprocal  $z^{-1} = \frac{1}{z}$ .

Find the reciprocal of  $z = 2 - 3i$ .

**SOLUTION:**

$$\frac{1}{z} = \frac{1}{2 - 3i} = \frac{1}{2 - 3i} \times \frac{2 + 3i}{2 + 3i} = \frac{2 + 3i}{2^2 + 3^2} = \frac{2}{13} + \frac{3}{13}i.$$

In what follows we discuss some results obtained when arithmetic operations are performed on complex numbers expressions involving conjugates.

Recall that for the complex number  $z = a + bi$ ,  $z\bar{z} = a^2 + b^2$  which is precisely the square of the modulus of  $z$ , i.e.  $|z|^2 = a^2 + b^2$ . We therefore obtain the following relation

$$|z|^2 = z\bar{z} \quad \text{and} \quad |z| = \sqrt{z\bar{z}}.$$

$$\arg(z_1 z_2) = \arg(z_1) + \arg(z_2) \quad \text{and} \quad \arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2). \quad (6.5)$$

We now prove some important formulae about properties of the modulus, argument and conjugation:-

### Proposition 6.1

The modulus, argument and conjugate functions satisfy the following properties. Let  $z_1$  and  $z_2$  be complex numbers.

- $|z_1 z_2| = |z_1| |z_2|$  and  $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$  if  $z_2 \neq 0$ .
- $|z_1 + z_2| \leq |z_1| + |z_2|$ ,
- $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$ ,  $\overline{z_1 - z_2} = \overline{z_1} - \overline{z_2}$
- $\overline{z_1 \cdot z_2} = \overline{z_1} \cdot \overline{z_2}$ ,  $\overline{\frac{z_1}{z_2}} = \frac{\overline{z_1}}{\overline{z_2}}$ ,  $\overline{\bar{z}} = z$ .
- $|z|^2 = z\bar{z}$  and  $|z| = \sqrt{z\bar{z}}$
- $\arg(z_1 z_2) = \arg z_1 + \arg z_2$  if  $z_1, z_2 \neq 0$ ,
- $\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$  if  $z_1, z_2 \neq 0$
- $\arg \bar{z} = -\arg z$ .

A selection of the above statements is proved here; the remaining ones are left as exercises.

**Proof:**  $|z_1 z_2| = |z_1| |z_2|$ .

Let  $z_1 = a + bi$  and  $z_2 = c + di$ . Then  $z_1 z_2 = (ac - bd) + (bc + ad)i$  so that

$$\begin{aligned}
 |z_1 z_2| &= \sqrt{(ac - bd)^2 + (bc + ad)^2} \\
 &= \sqrt{a^2 c^2 + b^2 d^2 + b^2 c^2 + a^2 d^2} \\
 &= \sqrt{(a^2 + b^2)(c^2 + d^2)} \\
 &= \sqrt{a^2 + b^2} \sqrt{c^2 + d^2} \\
 &= |z_1| |z_2|.
 \end{aligned}$$

**Proof:**  $\overline{z_1 \cdot z_2} = \overline{z_1} \cdot \overline{z_2}$

Let  $z_1 = a + bi$  and  $z_2 = c + di$ . Then

$$\begin{aligned}
 \overline{z_1 z_2} &= \overline{(ac - bd) + (bc + ad)i} \\
 &= (ac - bd) - (bc + ad)i \\
 &= (a - bi)(c - di) = \overline{z_1} \cdot \overline{z_2}
 \end{aligned}$$

**Proof:**  $\arg(z_1 z_2) = \arg z_1 + \arg z_2$  if  $z_1, z_2 \neq 0$ ,

Suppose  $z_1 = r_1 (\cos \theta_1 + i \sin \theta_1)$  and  $z_2 = r_2 (\cos \theta_2 + i \sin \theta_2)$ , then

$$\begin{aligned}
 z_1 z_2 &= r_1 r_2 (\cos \theta_1 + i \sin \theta_1) (\cos \theta_2 + i \sin \theta_2) \\
 &= r_1 r_2 \left[ (\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) \right. \\
 &\quad \left. + i (\cos \theta_2 \sin \theta_1 + \cos \theta_1 \sin \theta_2) \right] \\
 &= r_1 r_2 [\cos (\theta_1 + \theta_2) + i \sin (\theta_1 + \theta_2)].
 \end{aligned} \tag{6.6}$$

Proof:  $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$  if  $z_2 \neq 0$ ,

For  $z_2 \neq 0$ ,

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{r_1 (\cos \theta_1 + i \sin \theta_1)}{r_2 (\cos \theta_2 + i \sin \theta_2)} \\ &= \frac{r_1 (\cos \theta_1 + i \sin \theta_1)}{r_2 (\cos \theta_2 + i \sin \theta_2)} \cdot \frac{(\cos \theta_2 - i \sin \theta_2)}{(\cos \theta_2 - i \sin \theta_2)} \\ &= \frac{r_1 (\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2) + i (\cos \theta_2 \sin \theta_1 - \cos \theta_1 \sin \theta_2)}{r_2 (\cos^2 \theta_2 + \sin^2 \theta_2)} \quad (6.7) \\ &= \frac{r_1}{r_2} [\cos (\theta_1 - \theta_2) + i \sin (\theta_1 - \theta_2)]. \end{aligned}$$

It follows that

$$\left| \frac{z_1}{z_2} \right| = \frac{|r_1|}{|r_2|} = \frac{|r_1|}{|r_2|} = \frac{|z_1|}{|z_2|}. \quad (6.8)$$

### EXAMPLE 6.6

Let  $z_1 = i$  and  $z_2 = -\sqrt{3} - i$ . Then  $\text{Arg } (z_1) = \frac{\pi}{2}$  and  $\text{Arg } (z_2) = \frac{\pi}{2}$ . Thus from (6.1) we have

$$\begin{aligned} \arg (z_1 z_2) &= \frac{\pi}{2} + \left( -\frac{5\pi}{6} \right) = -\frac{\pi}{3} \quad \text{and} \\ &= \arg \left( \frac{z_1}{z_2} \right) = \frac{\pi}{2} - \left( -\frac{5\pi}{6} \right) = \frac{4\pi}{3}. \end{aligned}$$

### STRATEGY

In other words, given two complex numbers  $z_1$  and  $z_2$  to compute the arguments and modulus on their product/quotient proceed by computing arguments or modulus separately then combine the results via the relations on Proposition 6.1.

Find the modulus  $|z_1/z_2|$  when  $\begin{cases} z_1 = 1 + 2i \\ z_2 = 1 - 3i \end{cases}$

Clumsy method:

$$\begin{aligned} \left| \frac{z_1}{z_2} \right| &= \frac{|z_1 z_2^*|}{|z_2|^2} \\ &= \frac{|(1 + 2i)(1 + 3i)|}{1 + 9} \\ &= \frac{|(1 - 6) + i(2 + 3)|}{10} \\ &= \frac{\sqrt{25 + 25}}{10} = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}} \end{aligned}$$

Elegant method:

$$\begin{aligned} \left| \frac{z_1}{z_2} \right| &= \frac{|z_1|}{|z_2|} \\ &= \frac{\sqrt{1 + 4}}{\sqrt{1 + 9}} = \frac{1}{\sqrt{2}} \end{aligned}$$

### KEYPOINT

We have introduced complex numbers as a way to extend roots of polynomials and shown we can perform basic arithmetic with them. However, we can abstract and define a complex number system  $\mathbb{C}$  as an ordered pair  $(a, b)$  of real numbers, with addition and multiplication given by

$$(a, b) + (c, d) = (a + c, b + d)$$

$$(a, b) \cdot (c, d) = (ac - bd, ad + cb)$$

Now we have

$$(0, 1)(0, 1) = (-1, 0)$$

This will be route followed in a standard course in complex analysis. Notice we can recover the same operations on complex numbers if we care

to abbreviate  $(a, 0)$  to  $a$  and  $(0, 1)$  to  $i$ . Then we write

$$(a, b) = a + bi$$

and we have no ambiguity whatsoever.

### Exercises

1. Write the given complex number in polar form using the following argument representations

(i)  $\theta \neq \text{Arg}(z)$ .

(ii)  $\theta = \text{Arg}(z)$ .

(a)  $3$ .

(c)  $1 + i$ .

(b)  $4i$ .

(d)  $\frac{8}{\sqrt{2} + 2i}$ .

2. Use a calculator to write the given complex number in polar form the following argument representations

(i)  $\theta \neq \text{Arg}(z)$ .

(ii)  $\theta = \text{Arg}(z)$ .

(a)  $\sqrt{3} - \sqrt{5}i$ .

(c)  $-\sqrt{2} + 6i$ .

(b)  $2 + 3i$ .

(d)  $-5 - 3i$ .

3. Write the complex numbers whose polar coordinates  $(r, \theta)$  are given in the form  $a + ib$ . Use a calculator where necessary.

(a)  $\left(2, \frac{2\pi}{3}\right)$ .

(c)  $\left(1, -\frac{3\pi}{2}\right)$ .

(b)  $\left(-3, \frac{7\pi}{2}\right)$ .

(d)  $(3, 3)$ .

4. Write the complex numbers whose polar form are given in the form  $a + ib$ .

(a)  $z = 3 \left( \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$ .

(b)  $z = 2\sqrt{5} \left( \cos \frac{7\pi}{3} - i \sin \frac{7\pi}{3} \right)$ .

(c)  $z = 7 \left( \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$ .

(d)  $z = \sqrt{3} \left( \cos \frac{\pi}{5} - i \sin \frac{\pi}{5} \right)$ .

5. Evaluate the following powers of  $i$

(a)  $i^7$

(b)  $i^{40}$

(c)  $i^{13}$

(d)  $i^{105}$

6. Write the given numbers in the form  $a + ib$ .

(a)  $3i^4 - 4i^3 - 7i$

(b)  $\frac{3}{i^5} + \frac{i^3}{3} - \frac{20}{i^{18}}$

(c)  $2i^6 + \left( \frac{2}{-i} \right)^3 + 5i^{-5} - 12i$

(d)  $4i^7 + i^8 - 9i^3 + 10i^6 + 7$

(e)  $\left( \frac{1}{3} - \frac{i^5}{4} \right) \left( \frac{2}{3i^4} + \frac{5i^4}{3} \right)$

(f)  $3i + \frac{1}{2-i}$

(g)  $\frac{(1+i)(1-2i)}{(2+i)(4-3i)}$

(h)  $(1+i)^{10}$ ; Hint: Use binomial expansion.

7. Let  $z = x + iy$ . Determine the following expressing your answer in terms of  $x$  and  $y$ .

$$(a) \operatorname{Re} \left( \frac{2i}{z} \right)$$

$$(c) \operatorname{Im} (2\bar{z} + 4z - 3)$$

$$(b) \operatorname{Re} (z^3)$$

$$(d) \operatorname{Im} (\bar{z}^2 + z^2)$$

8. Let  $z = x + iy$ . Express the given quantity in terms of the symbols  $\operatorname{Re} (z)$  and  $\operatorname{Im} (z)$

$$(a) \operatorname{Re} (iz)$$

$$(c) \operatorname{Im} ((1 + i)z)$$

$$(b) \operatorname{Im} (iz)$$

$$(d) \operatorname{Re} (z^2)$$

9. Solve the following equations

$$(a) 2z = i(2 + 9i)$$

$$(b) z - 2\bar{z} + 7 - 6i = 0$$

$$(c) z^2 = i$$

$$(d) \bar{z}^2 = 4z$$

### 6.1.3 Inequalities

We note that order relations in  $\mathbb{R}$  does not hold in  $\mathbb{C}$ . So we cannot compare complex numbers by means of inequalities. But since the modulus of a complex number is a real number, we can define order relation on the modulus of two complex numbers.

For example, if  $z_1 = 3 + 4i$  and  $z_2 = 5 - i$ , then  $|z_1| = \sqrt{3^2 + 4^2} = 5$  and  $|z_2| = \sqrt{5^2 + (-1)^2} = \sqrt{26}$ . Hence  $|z_1| < |z_2|$ . This means that the complex number  $3 + 4i$  is nearer the origin than the complex number  $5 - i$ . As any two real numbers  $a, b$  satisfy the triangle  $|a + b| \leq |a| + |b|$ . The same inequality holds for a pair of complex numbers as follows:

$$|z_1 + z_2| \leq |z_1| + |z_2|.$$

Geometrically, the triangular inequality on complex numbers can be visualized as in Fig 6.6.

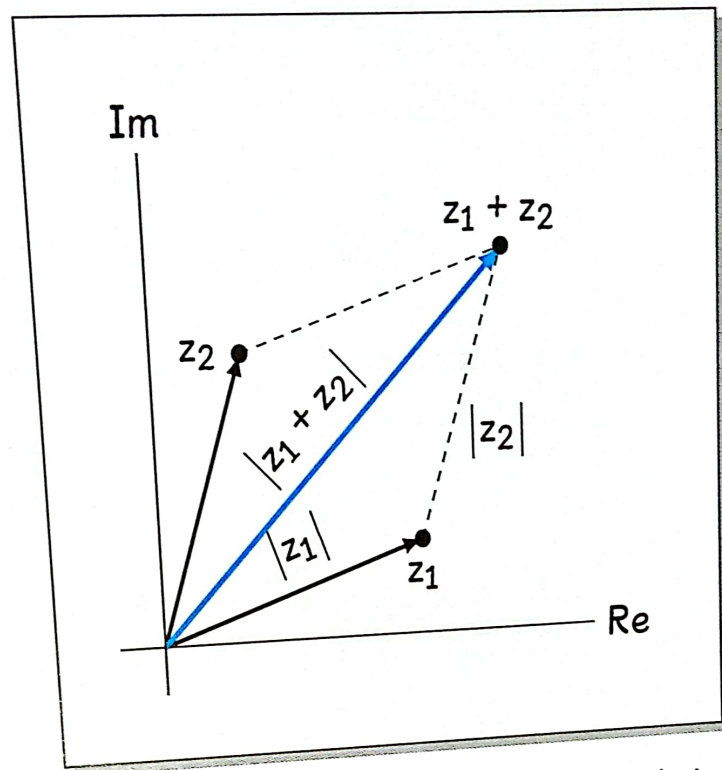


Figure 6.6: Diagrammatic proof of the triangle inequality

The triangle inequality can be extended to any finite sum of complex numbers; i.e.

$$|z_1 + z_2 + z_3 + \dots + z_n| \leq |z_1| + |z_2| + |z_3| + \dots + |z_n|. \quad (6.10)$$

By triangle inequality we obtain

$$|z_1| = |z_1 + z_2 + (-z_2)| \leq |z_1 + z_2| + |-z_2|.$$

Since  $|z_2| = |-z_2|$ , the above equation yields

$$|z_1| - |z_2| \leq |z_1 + z_2| \quad (6.11)$$

We see from the complex plane that the distance between the two points  $z_1 = x_1 + iy_1$  and  $z_2 = x_2 + iy_2$  is  $(x_2 - x_1, y_2 - y_1)$ . Therefore for  $z = z_2 - z_1$

$$\begin{aligned} |z| &= |z_2 - z_1| = |(x_2 + iy_2) - (x_1 + iy_1)| \\ &= |(x_2 - x_1) + i(y_2 - y_1)| \\ &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}. \end{aligned}$$

**EXAMPLE 6.8**

Determine the maximum value of  $\left| \frac{-1}{z^4 + 3z^2 + 2} \right|$  if  $|z| = 2$ .

**SOLUTION:**

From (6.1),

$$\left| \frac{-1}{z^4 + 3z^2 + 2} \right| = \frac{|-1|}{|z^4 + 3z^2 + 2|} = \frac{1}{|z^4 + 3z^2 + 2|}.$$

We seek a positive real number  $M$  such that

$$\frac{1}{|z^4 + 3z^2 + 2|} \leq M.$$

This is obtained by determining the least value of the denominator. Now  $z^4 + 3z^2 + 2 = (z^2 + 1)(z^2 + 2)$ . Thus

$$\begin{aligned} |z^4 + 3z^2 + 2| &= |z^2 + 1| |z^2 + 2| \\ &= |z^2 - (-1)| |z^2 - (-2)| \\ &\geq ||z^2| - |-1|| |z^2| - |-2|| \\ &= ||z^2| - 1| ||z^2| - 2| \end{aligned}$$

Since  $|z| = 2$ , it follows that

$$|z^4 + 3z^2 + 2| \geq |2^2 - 1| |2^2 - 2| = 3 \cdot 2 = 6.$$

Hence for  $|z| = 2$ , we have

$$\left| \frac{-1}{z^4 + 3z^2 + 2} \right| = \frac{1}{|z^4 + 3z^2 + 2|} \leq \frac{1}{6}.$$

**EXAMPLE 6.9**

Describe the set of points  $z$  in the complex plane that satisfy  $|z| = |z - i|$ .

**SOLUTION:**

This is the set of points which are equidistant from the origin and the point  $z = i$  since  $|z|$  is the distance from the origin to  $z$  and  $|z - i|$  is the distance

from  $z$  to  $i$ .

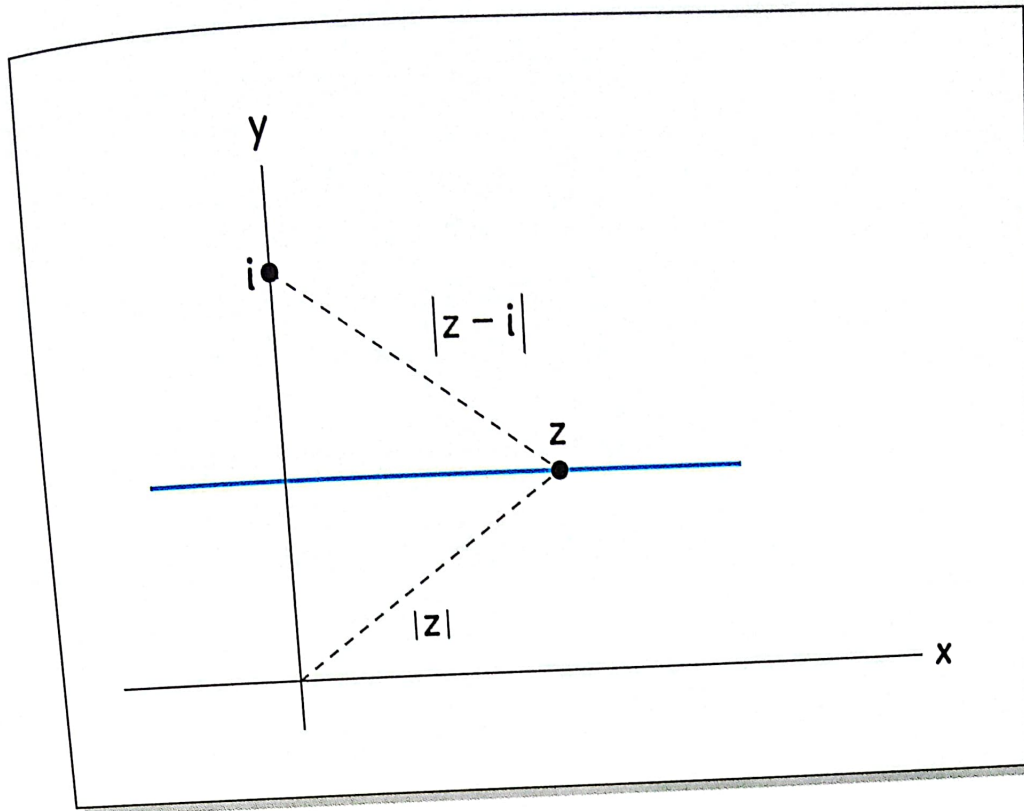


Figure 6.7

Analytically,  $|z| = |z - i|$  is the same as

$$\begin{aligned}\sqrt{x^2 + y^2} &= \sqrt{x^2 + (y - 1)^2} \\ x^2 + y^2 &= x^2 + y^2 - 2y + 1 \\ \Rightarrow y &= \frac{1}{2}.\end{aligned}$$

The equation holds for all  $x$  and so  $y = \frac{1}{2}$  is the equation of the horizontal line shown in color in Figure 6.7. The set of complex numbers satisfying  $|z| = |z - i|$  is therefore given by the equation  $z = x + \frac{1}{2}i$ .

1. In the following questions, interpret  $z_1$  and  $z_2$  as vectors. Graph  $z_1$ ,  $z_2$  and the indicated sum and differences as vectors.

(a)  $z_1 = 5 - 3i$ ,  $z_2 = -5 + 2i$ ,  $z_1 + z_2$ ,  $z_2 - z_1$ .

(b)  $z_1 = 3i$ ,  $z_2 = 1 - i$ ,  $3z_1 + 5z_2$ ,  $z_1 - 2z_2$ .

(c)  $z_1 = 5 + 4i$ ,  $z_2 = -3i$ ,  $2z_1 + 4z_2$ ,  $z_1 - z_2$ .

(d)  $z_1 = 3 + 4i, z_2 = 2 - 5i, 2z_1 + 4z_2, z_2 - z_1.$

2. (a) Plot the points  $z_1 = 3 + 4i, z_2 = -2i, z_3 = 6 + 3i.$

(b) Express each side of the triangle determined by the points in part (a) as a difference of vectors.

3. Find the modulus of the following complex numbers

(a)  $(2 + i)^2.$

(b)  $i(3 + i) + 2\left(\frac{1}{2} + \frac{i}{4}\right).$

(c)  $\frac{3 - 2i}{2 + 3i}.$

(d)  $\frac{2 + i}{1 - i} + \frac{3 - 2i}{1 + i}.$

4. In the following questions let  $z = x + iy$ . Express the given quantity in terms of  $x$  and  $y$

(a)  $|2z + 1 + 3i|^2.$

(c)  $|z^2 - 1 - 3i|.$

(b)  $|3z + \overline{z^2}|.$

(d)  $|z + 5\overline{z}|.$

5. In the following questions describe the set of points  $z$  in the complex plane that satisfy the given equation

(a)  $\operatorname{Re}((2 + i)z - i) = 1.$

(b)  $\left| \operatorname{Im}\left(i\left(\overline{z^2}\right)\right) \right| = 3.$

(c)  $|z + 2i| = |z - 3i|.$

$$(d) \operatorname{Re}(z^3) = |\sqrt{3} - i|.$$

6. Establish the given simultaneous inequalities

(a) If  $|z| = 2$ , then  $8 \leq |z + 6 + 8i| \leq 12$ .

(b) If  $|z| = 1$ , then  $2 \leq |z^2 - 3| \leq 4$ .

## 6.2 Elementary functions of complex variable

This section is devoted to basic functions of complex variable and simple applications. We give de Moivre's theorem and show examples of its uses. We end by discussing applications of complex variable to finding roots of polynomials.

### EXAMPLE 6.10

Consider the complex function  $f : \mathbb{C} \rightarrow \mathbb{C}$  given by  $f(z) = z^2$ . Separating real and imaginary parts,  $z = x + iy$ , we have

$$f(z) = z^2 = (x + iy)(x + iy) = x^2 - y^2 + 2ixy$$

Similarly,

$$f(z) = \frac{1}{z} = \frac{z^*}{|z|^2} = \frac{x}{x^2 + y^2} - i \frac{y}{x^2 + y^2}.$$

### 6.2.1 The complex exponential function

The definition of the exponential, cosine and sine functions of a **real** variable can be done by writing their series expansions:

$$\begin{aligned} e^x &= 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \dots \\ \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + (-1)^n \frac{x^{2n}}{(2n)!} + \dots \\ \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \dots \end{aligned} \tag{6.12}$$



For a small  $x$  a few of terms may be sufficient to provide a good approximation. Thus for very small  $x$ ,

$$\sin x \approx x.$$

In a similar manner we may define functions of the complex variable  $z$ . The complex exponential is defined by

$$e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} \dots \quad (6.13)$$

A special case is if  $z$  is purely imaginary  $z = i\theta$  we obtain the Euler's formula. Using the fact that  $i^{2n} = 1$  or  $-1$  for  $n$  even or odd and  $i^{2n+1} = i$  or  $-i$  for  $n$  even or odd

$$\begin{aligned} e^{i\theta} &= 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} \dots \\ &= 1 + i\theta - \frac{\theta^2}{2!} - i\frac{\theta^3}{3!} + \frac{\theta^4}{4!} \dots \\ &= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} + \dots\right) + i\left(\theta - \frac{\theta^3}{3!} + \dots\right) \\ &= \cos \theta + i \sin \theta \end{aligned} \quad (6.14)$$

The upshot of this is the relations of a complex number in polar form now to a representation in exponential form. Recall writing a complex number in polar form,

$$\begin{aligned} z &= |z|(\cos \theta + i \sin \theta) = r(\cos \theta + i \sin \theta) = re^{i\theta} \\ z^* &= |z|(\cos \theta - i \sin \theta) = r(\cos \theta - i \sin \theta) = re^{-i\theta} \end{aligned} \quad (6.15)$$

Also note that

$$\frac{1}{z} = \frac{z^*}{zz^*} = \frac{|z|(\cos \theta - i \sin \theta)}{|z|^2} = \frac{e^{-i\theta}}{|z|} = \frac{1}{r}e^{-i\theta}.$$

We may find a useful relation between sines and cosines and complex exponentials from the fact that

$$\begin{aligned} e^{i\theta} &= \cos \theta + i \sin \theta \\ e^{-i\theta} &= \cos \theta - i \sin \theta. \end{aligned} \quad (6.16)$$

Adding and then subtracting the two of equations (6.16) we find that

$$\begin{aligned}\cos \theta &= \frac{1}{2}(e^{i\theta} + e^{-i\theta}) \\ \sin \theta &= \frac{1}{2i}(e^{i\theta} - e^{-i\theta})\end{aligned}\tag{6.17}$$

## The complex sine and cosine functions

In a similar manner we can define  $\cos z$  and  $\sin z$  by replacing the argument  $x$  in equations (6.12) by the complex variable  $z$ . The analogue of de Moivre's theorem is

$$\begin{aligned}e^{iz} &= \left(1 - \frac{z^2}{2!} + \frac{z^4}{4!} + \dots\right) + i\left(z - \frac{z^3}{3!} + \dots\right) \\ &= \cos z + i \sin z\end{aligned}$$

Similarly one has

$$\begin{aligned}\cos z &= \frac{1}{2}(e^{iz} + e^{-iz}) \\ \sin z &= \frac{1}{2i}(e^{iz} - e^{-iz})\end{aligned}\tag{6.18}$$

From this we learn that the cosine and the sine of an imaginary angle are

$$\begin{aligned}\cos(ib) &= \frac{1}{2}(e^{-b} + e^b) = \cosh b \\ \sin(ib) &= \frac{1}{2i}(e^{-b} - e^b) = i \sinh b\end{aligned}\tag{6.19}$$

where we have used the definitions of the hyperbolic functions

$$\begin{aligned}\cosh b &\equiv \frac{1}{2}(e^b + e^{-b}) \\ \sinh b &\equiv \frac{1}{2}(e^b - e^{-b})\end{aligned}\tag{6.20}$$



Hyperbolic functions get their name from the identity

$$\cosh^2 \theta - \sinh^2 \theta = 1,$$

which is readily proved from (6.20) and is reminiscent of the equation of a hyperbola,

$$x^2 - y^2 = 1.$$

## 6.2.2 de Moivre's theorem and trigonometric identities

If  $z = a + bi = re^{i\theta} = r(\cos \theta + i \sin \theta)$  then for any integer  $n$  using the rules for multiplication of complex numbers we obtain de Moivre's theorem:

$$z^n = (re^{i\theta})^n = r^n e^{in\theta} = r^n (\cos n\theta + i \sin n\theta) \quad (6.21)$$

That is,

$$(e^{i\theta})^n = (\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta. \quad (6.22)$$

This result is known as de Moivre's formula.

### Uses of de Moivre's theorem

This equation (6.22) generates simple identities for  $\cos n\theta$  and  $\sin n\theta$ .

#### EXAMPLE 6.11

Let  $n = 2$

$$(\cos \theta + i \sin \theta)^2 = \cos 2\theta + i \sin 2\theta.$$

de Moivre's theorem

$$(\cos \theta + i \sin \theta)^2 = \cos^2 \theta + i2 \cos \theta \sin \theta - i \sin^2 \theta \quad \text{by binomial theorem}$$

equating the real and imaginary parts of the equation, we have

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$\sin 2\theta = 2 \cos \theta \sin \theta$$

In general the complex exponential is very useful in establishing trigonometric identities.

#### EXAMPLE 6.12

Deduce that

$$\cos(a + b) = \cos a \cos b - \sin a \sin b$$

$$\sin(a + b) = \sin a \cos b + \cos a \sin b$$

SOLUTION:

$$e^{i(a+b)} = \cos(a+b) + i \sin(a+b)$$

$$\begin{aligned} e^{ia} e^{ib} &= (\cos a + i \sin a)(\cos b + i \sin b) \\ &= (\cos a \cos b - \sin a \sin b) + i(\cos a \sin b + \sin a \cos b) \end{aligned}$$

where we have used the property of exponentials that  $e^{i(a+b)} = e^{ia} e^{ib}$ . In particular we have established that:

$$\cos(a+b) + i \sin(a+b) = (\cos a \cos b - \sin a \sin b) + i(\cos a \sin b + \sin a \cos b)$$

Comparing real and imaginary parts of the two sides we deduce that

$$\begin{aligned} \cos(a+b) &= \cos a \cos b - \sin a \sin b \\ \sin(a+b) &= \sin a \cos b + \cos a \sin b \end{aligned}$$

### EXAMPLE 6.13

Compute  $z^3$  for  $z = -\sqrt{3} - i$ .

#### SOLUTION:

Here  $r = 2$ ,  $\arg(z) = \frac{7\pi}{6}$  and  $n = 3$ . Thus

$$\begin{aligned} (-\sqrt{3} - i)^3 &= 2^3 \left[ \cos\left(3 \cdot \frac{7\pi}{6}\right) + i \sin\left(3 \cdot \frac{7\pi}{6}\right) \right] \\ &= 8 \left[ \cos \frac{7\pi}{2} + i \sin \frac{7\pi}{2} \right] \\ &= 8 \left[ \cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right] = 8 \cdot (-i) = -8i. \end{aligned}$$

### EXAMPLE 6.14

Compute  $\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^3$ .

#### SOLUTION:

Here  $r = \sqrt{\left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = 1$ ,  $\arg(z) = \tan^{-1} \frac{1}{\sqrt{3}} = \frac{\pi}{6}$ . Thus

$$\begin{aligned}\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^3 &= \cos 3\left(\frac{\pi}{6}\right) + i \sin 3\left(\frac{\pi}{6}\right) \\ &= \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} = i.\end{aligned}$$

As illustrated in the following example, De Moivre's formula can be used to establish a number of trigonometric identities

### EXAMPLE 6.15

Prove the identities

(a)  $\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$ .

(b)  $\frac{\sin 5\theta}{\sin \theta} = 16 \sin^4 \theta - 12 \cos^2 \theta + 1$  if  $\theta \neq 0, \pm\pi, \pm2\pi, \dots$

#### SOLUTION:

We use the binomial formula

$$(a + b)^5 = \sum_{k=1}^5 \frac{5!}{k! (5-k)!} a^{5-k} b^k.$$

Hence with the identification  $a = \cos \theta$  and  $b = \sin \theta$  we have

$$\begin{aligned}\cos 5\theta + i \sin 5\theta &= (\cos \theta + i \sin \theta)^5 \\ &= \cos^5 \theta + 5 (\cos^4 \theta) (i \sin \theta) + 10 (\cos^3 \theta) (i \sin \theta)^2 \\ &\quad + 10 (\cos^2 \theta) (i \sin \theta)^3 + 5 (\cos \theta) (i \sin \theta)^4 + (i \sin \theta)^5 \\ &= \cos^5 \theta + 5i \cos^4 \theta \sin \theta - 10 \cos^3 \theta \sin^2 \theta \\ &\quad - 10i \cos^2 \theta \sin^3 \theta + 5 \cos \theta \sin^4 \theta + i \sin^5 \theta \\ &= \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta \\ &\quad + i (5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta).\end{aligned}$$

Therefore

$$(a) \cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta$$

$$= \cos^5 \theta - 10 \cos^3 \theta (1 - \cos^2 \theta) + 5 \cos \theta (1 - \cos^2 \theta)^2$$

$$= 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta.$$

and,

$$(b) \frac{\sin 5\theta}{\sin \theta} = 5 \cos^4 \theta - 10 \cos^2 \theta \sin^2 \theta + \sin^4 \theta$$

$$= 5 \cos^4 \theta - 10 \cos^2 \theta (1 - \cos^2 \theta) + (1 - \cos^2 \theta)^2$$

$$= 16 \cos^4 \theta - 12 \cos^2 \theta + 1 \text{ provided } \theta \neq 0, \pm\pi, \pm2\pi, \dots$$



It is often much easier and more compact to work with the complex exponential rather than with sines and cosines. Here is an example:

#### EXAMPLE 6.16

Find  $(1 + i)^8$ . Taking powers is much simpler in polar form so we write  $(1 + i) = \sqrt{2}e^{i\pi/4}$ . Hence  $(1 + i)^8 = (\sqrt{2}e^{i\pi/4})^8 = 16e^{2\pi i} = 16$ .

### 6.2.3 Complex logarithm

The logarithmic function  $f(z) = \ln z$  is the inverse of the exponential function meaning that if one acts on  $z$  by the logarithmic function and then by the exponential function one gets just  $z$ ,  $e^{\ln z} = z$ . We may use this property to define the logarithm of a complex variable:

$$e^{\ln z} = z = |z|e^{i\theta} = e^{\ln |z|}e^{i\theta} = e^{\ln |z| + i\theta} \Rightarrow \ln z = \ln |z| + i \arg(z) = a + ib. \quad (6.23)$$



Note that the infinite ambiguity in the phase of  $z$  is no longer present in  $\ln z$  because the addition of an integer multiple of  $2\pi$  to the argument of  $z$  changes the imaginary part of the logarithm by the same amount. Thus it is essential, when defining the logarithm, to know precisely the argument of  $z$ .

We can rewrite equation (6.23) explicitly as

$$\ln z = \ln |z| + i(\theta + 2\pi n), \quad n \text{ integer.} \quad (6.24)$$

For different  $n$  we get different values of the complex logarithm. So we need to assign  $n$  to fully specify the logarithm.

The different values corresponding to different  $n$  are called "branches" of the logarithm.  $n = 0$  is called the **principal branch**.

A "function" of  $z$  which may take not one but multiple values for a given value of  $z$  is called multi-valued. The logarithm is our first example of a multi-valued function.

### EXAMPLE 6.17

Find all values of  $\ln(-1)$ .

**SOLUTION:**

$$\ln(-1) = \ln e^{i\pi} = \ln 1 + i(\pi + 2\pi n) = i\pi + 2\pi in, \quad n \text{ integer.}$$

For the principal branch  $n = 0$

$$\ln(-1) = i\pi.$$



Note:  $e^{\ln z}$  always equals  $z$ , while  $\ln e^z$  does not always equal  $z$ . Let  $z = a + ib = re^{i\theta}$ . Then  $\ln z = \ln r + i(\theta + 2\pi n)$ ,  $n$  integer.

So

$$e^{\ln z} = e^{r+i(\theta+2\pi n)} = re^{i\theta} \underbrace{e^{2\pi ni}}_1 = re^{i\theta} = z.$$

On the other hand  $e^z = e^{a+ib} = e^a e^{ib}$ . Therefore

$$\ln e^z = \ln e^a + i(b + 2\pi n) = \underbrace{a + ib}_z + 2\pi in = z + 2\pi in \neq z.$$

## Complex powers

Once we have the complex logarithm, we can define complex powers  $f(z) = z^\alpha$ , where both  $z$  and  $\alpha$  are complex:

$$f(z) = z^\alpha = e^{\alpha \ln z}. \quad (6.25)$$

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Since the logarithm is multi-valued, complex powers also are multi-valued functions.

EXAMPLE 6.18

Show that  $i^i$  is real and the principal-branch value is  $i^i = 1/\sqrt{e^\pi}$ .

SOLUTION:

$$\begin{aligned} i^i &= e^{i \ln i} = e^{i[\ln 1 + i(\pi/2 + 2\pi n)]} \\ &= e^{-\pi/2 - 2\pi n} \end{aligned}$$

These values are all real. For  $n = 0$  we have

$$i^i = e^{-\pi/2} = 1/\sqrt{e^\pi}.$$

### 6.2.4 Roots of complex numbers

We note that the function  $f(z) = z^{1/n}$  is a multi-valued function.

**DEFINITION 6.10**

A number  $w$  is defined to be an  $n$ -th root of complex number  $z$  if  $w^n = z$ . Then we write  $w = z^{1/n}$ .

To find powers and root of complex numbers it is almost always easiest to write them as complex exponentials. The following result obtains.

#### Proposition 6.2

Every complex number has exactly  $n$  distinct  $n$ -th roots.

**Proof:** Let  $z = r(\cos \theta + i \sin \theta)$ ;  $w = \rho(\cos \alpha + i \sin \alpha)$ . Then, using de Moivre's theorem,

$$\begin{aligned} r(\cos \theta + i \sin \theta) &= \rho^n (\cos \alpha + i \sin \alpha)^n \\ &= \rho^n (\cos n\alpha + i \sin n\alpha) \\ \Rightarrow \rho^n &= r, \quad n\alpha = \theta + 2\pi k \quad (k \text{ integer}) \end{aligned}$$

Therefore  $\rho = r^{1/n}$ ,  $\alpha = \theta/n + 2\pi k/n$ .

We thus see that we get  $n$  distinct values for  $k$ , from  $0$  to  $n-1$ , corresponding to  $n$  distinct values of  $w$  for  $z \neq 0$ .

In summary the  $n^{\text{th}}$  roots of a nonzero complex number

$$z = r (\cos \theta + i \sin \theta)$$

are given by

$$w_k = \sqrt[n]{r} \left[ \cos \left( \frac{\theta + 2k\pi}{n} \right) + i \sin \left( \frac{\theta + 2k\pi}{n} \right) \right] \quad (6.26)$$

where  $k = 0, 1, 2, \dots, n-1$ . ■

### KEYPOINT

Note that finding the  $n^{\text{th}}$  roots of a nonzero complex number is equivalent to solving a polynomial equation

$$x^n = z \quad (6.27)$$

which means we need to find  $w_k = z^{1/n}$  satisfying the polynomial equation (6.27).

Find the fourth roots of  $z = 1 + i$ .

**SOLUTION:**

Here  $r = \sqrt{2}$  and  $\arg(z) = \frac{\pi}{4}$ . With  $n = 4$ , we obtain

$$w_k = \sqrt[8]{2} \left[ \cos \left( \frac{\frac{\pi}{4} + 2k\pi}{4} \right) + i \sin \left( \frac{\frac{\pi}{4} + 2k\pi}{4} \right) \right], \quad k = 0, 1, 2, 3.$$

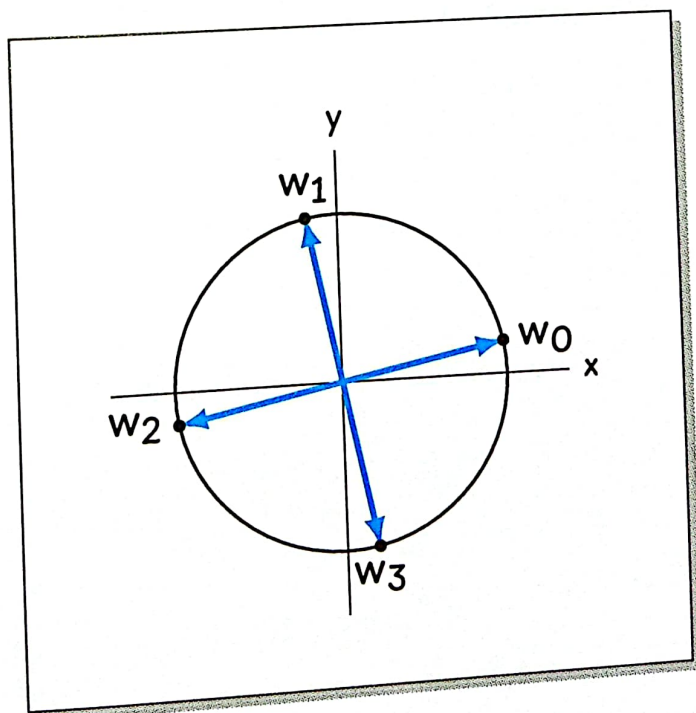
Thus

$$w_0 = \sqrt[8]{2} \left( \cos \frac{\pi}{16} + i \sin \frac{\pi}{16} \right),$$

$$w_1 = \sqrt[8]{2} \left( \cos \frac{9\pi}{16} + i \sin \frac{9\pi}{16} \right),$$

$$w_2 = \sqrt[8]{2} \left( \cos \frac{17\pi}{16} + i \sin \frac{17\pi}{16} \right),$$

$$w_3 = \sqrt[8]{2} \left( \cos \frac{25\pi}{16} + i \sin \frac{25\pi}{16} \right).$$



The four roots lie on a circle centered at the origin of radius  $r = \sqrt[8]{2}$  and at equal angular interval of  $\frac{2\pi}{4} = \frac{\pi}{2}$  radians beginning with the root whose argument is  $\frac{\pi}{16}$ .

# Complex Numbers

## EXAMPLE 6.20

Find all cubic roots of  $z = -1 + i$ .

### SOLUTION:

Applying equation (6.26) we find

$$w = (-1 + i)^{1/3}$$

$$w_k = (\sqrt{2})^{1/3} \left[ \cos\left(\frac{3\pi}{4} \frac{1}{3} + \frac{2\pi k}{3}\right) + i \sin\left(\frac{3\pi}{4} \frac{1}{3} + \frac{2\pi k}{3}\right) \right], \quad k = 0, 1, 2.$$

That is, the three cubic roots of  $-1 + i$  are

$$w_0 = 2^{1/6} \left( \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

$$w_1 = 2^{1/6} \left( \cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right)$$

$$w_2 = 2^{1/6} \left( \cos \frac{19\pi}{12} + i \sin \frac{19\pi}{12} \right).$$

Equivalently, using equation (6.25),

$$\begin{aligned} w &= (-1 + i)^{1/3} = e^{(1/3) \ln(-1+i)} \\ &= e^{(1/3)[\ln \sqrt{2} + i(3\pi/4 + 2k\pi)]} \\ &= (\sqrt{2})^{1/3} e^{i(\pi/4 + 2k\pi/3)} \\ &= 2^{1/6} e^{i(\pi/4 + 2k\pi/3)}. \end{aligned}$$

## EXAMPLE 6.21

Consider the equation

$$z^5 + 32 = 0$$

This is similar to the previous case. The solutions are the fifth roots of  $-32$ : we compute  $w = (-32)^{1/5}$

$$w_k = 32^{1/5} \left[ \cos \left( \frac{\pi}{5} + \frac{2\pi k}{5} \right) + i \sin \left( \frac{\pi}{5} + \frac{2\pi k}{5} \right) \right], \quad k = 0, 1, 2, 3, 4.$$

That is,

$$w_0 = 2 \left( \cos \frac{\pi}{5} + i \sin \frac{\pi}{5} \right)$$

$$w_1 = 2 \left( \cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5} \right)$$

$$w_2 = -2$$

$$w_3 = 2 \left( \cos \frac{7\pi}{5} + i \sin \frac{7\pi}{5} \right)$$

$$w_4 = 2 \left( \cos \frac{9\pi}{5} + i \sin \frac{9\pi}{5} \right)$$

## 6.2.5 The roots of unity

Consider the equation

$$x^n = 1$$

The solution is given by the  $n$ -th roots of unity  $\Rightarrow w = 1^{1/n}$ .

$$1 = e^{2k\pi i} \Rightarrow 1^{1/n} = e^{2k\pi i/n}, \quad k = 1, 2, 3, \dots, n-1$$

$$= \cos\left(\frac{2k\pi}{n}\right) + i \sin\left(\frac{2k\pi}{n}\right)$$

For example take  $n = 3$  the roots of  $z^3 = 1$  are

$$w_0 = 1$$

$$w_1 = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} = \frac{-1 + \sqrt{3}i}{2}$$

$$w_2 = \cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} = \frac{-1 - \sqrt{3}i}{2}$$

They lie at the vertices of a regular triangle on the unit circle in the complex plane.

For instance, if  $n = 5$ :

$$1^{1/5} = \cos\left(\frac{2k\pi}{5}\right) + i \sin\left(\frac{2k\pi}{5}\right) \quad (k = 0, 1, 2, 3, 4).$$

The roots may be drawn in the Argand plane and correspond to five equally spaced points in the plane:

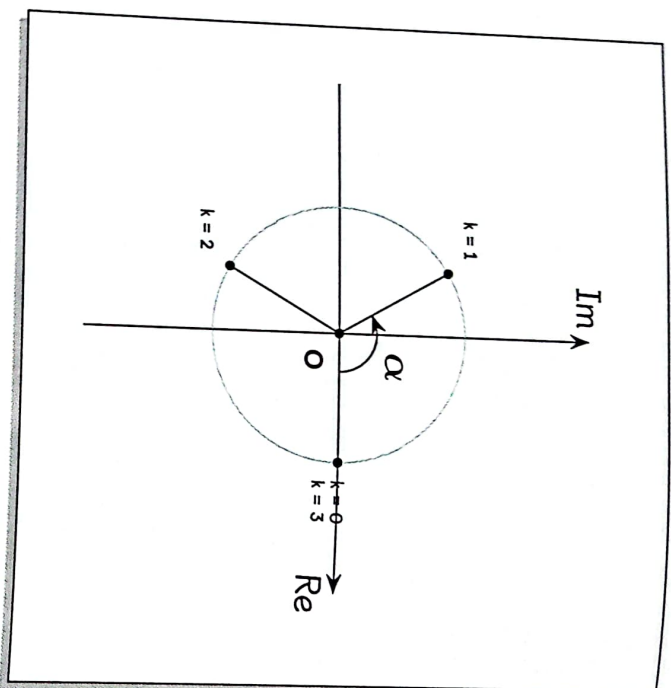


Figure 6.8

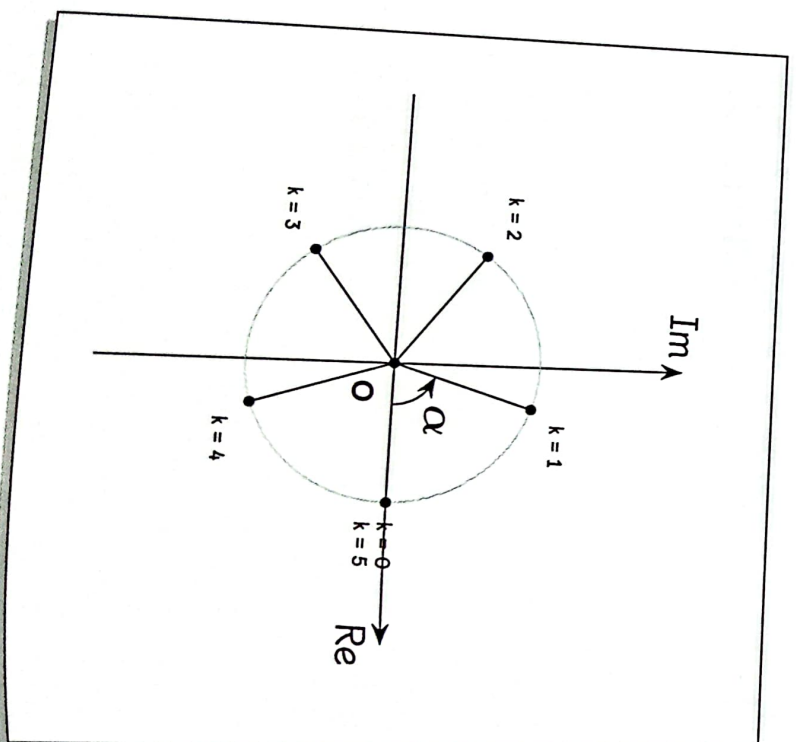


Figure 6.9

## KEYPOINT

In general the  $n$ -th roots of unity lie at the vertices of a regular  $n$ -gon.

## 6.3 Roots of Polynomials

In this section we start by defining and evaluating roots of complex numbers. Then we discuss how to characterize polynomials by their roots. Finally we give examples of how to use complex variable techniques to solve polynomial equations in special cases.

Complex numbers enable us to find roots for any polynomial

### DEFINITION 6.11

Let  $P(z)$  be polynomial of degree  $n$  in the complex variable  $z$

$$P(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_0. \quad (6.28)$$

where  $a_k$  are complex coefficients.  $r_i$  is root of  $P(z)$  if  $P(r_i) = 0$ .

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If  $z$  is a root of the polynomial

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_0 = 0$$

where the  $a_i$  are real then so is its conjugate  $\bar{z}$ .

For instance, if a real quadratic equation  $ax^2 + bx + c = 0$  has complex roots then these roots are conjugates of each other.

### EXAMPLE 6.22

$$\begin{aligned} x^2 + 1 &= (x + i)(x - i) && \text{over } \mathbb{C} \\ x^3 - x^2 + 2x - 2 &= (x - 1)(x + i\sqrt{2})(x - i\sqrt{2}) && \text{over } \mathbb{C} \end{aligned}$$

The condition requiring that the polynomial has real coefficients is necessary for the roots to be complex conjugates of each other. We present the following examples because this is a common early misconception involving complex numbers.

**EXAMPLE 6.23**

Find the square root of  $i$ . In other words find all  $z$  that satisfy  $z^2 = i$ .

**SOLUTION:**

Suppose that  $z^2 = i$  and  $z = a + bi$ , where  $a$  and  $b$  are real. Then

$$i = (a + bi)^2 = (a^2 - b^2) + 2abi.$$

Comparing the real and imaginary parts we see that

$$a^2 - b^2 = 0$$

$$2ab = 1$$

So  $b = \pm a$  from the first equation. Substituting  $b = a$  into the second equation gives  $a = b = 1/\sqrt{2}$  or  $a = b = -1/\sqrt{2}$ .

Substituting  $b = -a$  into the second equation gives  $-2a^2 = 1$  which has no real solution in  $a$ . So the two  $z$  which satisfy  $z^2 = i$ , i.e. the two square roots of  $i$ , are

$$\frac{1+i}{\sqrt{2}} \text{ and } \frac{-1-i}{\sqrt{2}}.$$

Notice, as with square roots of real numbers, that the two square roots are negative one to each other and not complex conjugates.

### 6.3.1 Quadratic Polynomials

Consider the quadratic equation  $ax^2 + bx + c = 0$ . Recall the discriminant  $\Delta = b^2 - 4ac$  of the quadratic equation or polynomial. The nature of the roots of the quadratic depends on the sign of the discriminant. In particular,

(a) If  $\Delta > 0$ , then  $\Delta = k^2$  for some real  $k$ , and the quadratic equation has two distinct real roots  $\frac{-b+k}{2a}$  and  $\frac{-b-k}{2a}$ .

(b) If  $\Delta = 0$ , then the quadratic has one real root  $\frac{-b}{2a}$ .

(c) If  $\Delta < 0$ , then  $\Delta = -k^2$  for some real  $k$ , and the quadratic has two complex conjugate roots  $\frac{-b+ik}{2a}$  and  $\frac{-b-ik}{2a}$ .

Find the roots of these quadratic polynomials and thereby factor the polynomials into linear factors.

(a)  $x^2 + x - 1$

(b)  $9x^2 - 6x + 1$

(c)  $2x^2 + x + 1$

**SOLUTION:**

We use the quadratic formula to find the roots of these polynomials.

(a) Here  $A = 1, B = 1, C = -1$ , therefore

$$x = \frac{-1 \pm \sqrt{1+4}}{2} = -\frac{1}{2} \pm \frac{\sqrt{5}}{2},$$

and so

$$x^2 + x - 1 = \left(x + \frac{1}{2} - \frac{\sqrt{5}}{2}\right) \left(x + \frac{1}{2} + \frac{\sqrt{5}}{2}\right).$$

(b) Here  $A = 9, B = -6, C = 1$ , therefore

$$x = \frac{6 \pm \sqrt{36 - 36}}{18} = \frac{1}{3} \quad (\text{a double root})$$

and so

$$9x^2 - 6x + 1 = 9 \left(x^2 - \frac{2}{3}x + \frac{1}{9}\right) = 9 \left(x - \frac{1}{3}\right)^2 = (3x - 1)^2.$$

(c) Here  $A = 2, B = 1, C = 1$ , therefore

$$x = \frac{-1 \pm \sqrt{1-8}}{4} = -\frac{1}{4} \pm \frac{\sqrt{7}}{4}i,$$

# Complex Numbers

and so

$$\begin{aligned}2x^2 + x + 1 &= 2 \left( x^2 + \frac{1}{2}x + \frac{1}{2} \right) \\&= 2 \left( x + \frac{1}{4} - \frac{\sqrt{7}}{4}i \right) \left( x + \frac{1}{4} + \frac{\sqrt{7}}{4}i \right).\end{aligned}$$

Some quadratic and higher degree polynomials can be (at least partially) factored as indicated below;

(a) Difference of squares:  $x^2 - a^2 = (x - a)(x + a)$ .

(b) Difference of cubes:  $x^3 - a^3 = (x - a)(x^2 + ax + a^2)$ .

(c) More generally, a difference of  $n$ th powers for any positive integer  $n$ :

$$x^n - a^n = (x - a)(x^{n-1} + ax^{n-2} + a^2x^{n-3} + \dots + a^{n-2}x + a^{n-1}).$$

Note that  $x - a$  is a factor of  $x^n - a^n$  for any positive integer  $n$ .

(d) When  $n$  is an odd positive integer, then  $x + a$  is a factor of  $x^n + a^n$ . For example

$$x^3 + a^3 = (x + a)(x^2 - ax + a^2).$$

## EXAMPLE 6.25

Use the quadratic formula to find the two solutions of

$$z^2 - (3 + i)z + (2 + i) = 0.$$

### SOLUTION:

We see that  $a = 1$ ,  $b = -3 - i$ , and  $c = 2 + i$ . So

$$b^2 - 4ac = (-3 - i)^2 - 4 \times 1 \times (2 + i) = 9 - 1 + 6i - 8 - 4i = 2i.$$

Knowing

$$\sqrt{i} = \pm \frac{1+i}{\sqrt{2}}, \quad \text{from the previous problem,}$$

we have

$$\begin{aligned}x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{(3+i) \pm \sqrt{2i}}{2} = \frac{(3+i) \pm \sqrt{2}\sqrt{i}}{2} \\&= \frac{(3+i) \pm (1+i)}{2} = \frac{4+2i}{2} \text{ or } \frac{2}{2} = 2+i \text{ or } 1.\end{aligned}$$

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Again note that the two roots are not conjugates of one another – this need not be the case here as the coefficients  $a, b, c$  are not all real.

From the quadratic formula (6.1) we know that all quadratic equations ( $n = 2$ ) can be solved using complex numbers. Gauss was the first to prove the much more general result:

### THEOREM

6.1

### FUNDAMENTAL THEOREM OF ALGEBRA

The roots of any polynomial equation

$$a_0 + a_1x + a_2x^2 + \dots + a_nx^n = 0$$

with real or complex coefficients has  $n$  complex roots (not necessarily distinct). That is there are  $n$  complex numbers  $r_1, \dots, r_n$  such that

$$a_0 + a_1x + a_2x^2 + \dots + a_nx^n = a_n(x - r_1)(x - r_2) \dots (x - r_n).$$

In particular, an  $n$  degree polynomial has, counting multiplicities,  $n$  roots in  $\mathbb{C}$ .

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The proof of this theorem is far beyond the scope of this book. Note that the theorem only guarantees the **existence** of the roots of a polynomial somewhere in  $\mathbb{C}$  unlike the quadratic formula which plainly gives us the roots. There is a similar formula for the case  $n = 3$ . It can be shown that such general formulae do not exist for equations of higher order.

### KEYPOINT

The theorem gives no hints as to where in  $\mathbb{C}$  these roots are to be found. We can, however, find the roots of specially simple polynomials.

### EXAMPLE 6.26

What is the degree of  $P(x) = x^3(x^2 + 2x + 5)^2$ ? What are the roots of  $P$  and, what is the multiplicity of each root?

#### SOLUTION:

If  $P$  is expanded, the highest power of  $x$  present in the expansion is  $x^3(x^2)^2 = x^7$ , so  $P$  has degree 7. The factor  $x^3 = (x - 0)^3$  indicates that  $0$  is a root of  $P$  having multiplicity 3. The remaining two roots of  $x^2 + 2x + 5$ , each having multiplicity 2. Now

$$\begin{aligned}(x^2 + 2x + 5)^2 &= ((x + 1)^2 + 4)^2 \\ &= ((x + 1 + 2i)(x + 1 - 2i))^2.\end{aligned}$$

Hence the seven roots of  $P$  are:

$$\begin{cases} 0, 0, 0 & 0 \text{ has multiplicity } 3 \\ -1 - 2i, -1 - 2i & -1 - 2i \text{ has multiplicity } 2 \\ -1 + 2i, -1 + 2i & -1 + 2i \text{ has multiplicity } 2. \end{cases}$$

### 6.3.2 Characterizing a polynomial by its roots

For  $n = 2$  we have the quadratic formula for roots

$$r_{1,2} = -\frac{a_1}{2a_2} \pm \frac{1}{2a_2} \sqrt{a_1^2 - 4a_0a_2}.$$

The sum and product of the roots are given respectively by  
sum of roots  $= r_1 + r_2 = -\frac{a_1}{a_2}$ , product of roots  $= r_1 \cdot r_2 = \frac{a_0}{a_2}$ .

Here we use this result to notice that we can characterize  $P(z)$  equivalently by its coefficients or by its roots. More precisely,

## Proposition 6.3

A polynomial equation

$$P(z) = a_n z^n + a_{n-1} z^{n-1} + a_{n-2} z^{n-2} + \dots + a_0 = 0$$

has  $n$  roots say  $r_1, r_2, \dots, r_n$ , can be expressed as a product of linear terms.

$$\begin{aligned} a_n z^n + a_{n-1} z^{n-1} + \dots + a_0 &= a_n (z - r_1)(z - r_2) \dots (z - r_n) \\ &= a_n \left( z^n - z^{n-1} \sum_{j=1}^n r_j + \dots + (-1)^n \prod_{j=1}^n r_j \right). \end{aligned}$$

Comparing the coefficients of  $z^{n-1}$  and  $z^0$  on the two sides, we deduce that

$$\frac{a_{n-1}}{a_n} = - \sum_j r_j; \quad \frac{a_0}{a_n} = (-1)^n \prod_j r_j \quad (6.29)$$

the case for  $n = 2$  is easily seen.

Demonstration for  $n = 4$ :  $\prod_{i=1}^n r_i = (-1)^n a_0 / a_n$ .

$$\begin{aligned} (z - r_1)(z - r_2)(z - r_3)(z - r_4) &= z^4 - (r_1 + r_2 + r_3 + r_4)z^3 \\ &\quad + (r_1 r_2 + r_1 r_3 + r_1 r_4 + r_2 r_3 + r_2 r_4 + r_3 r_4)z^2 \\ &\quad - (r_1 r_2 r_3 + r_2 r_3 r_4 + r_3 r_4 r_1 + r_4 r_1 r_2)z + (r_1 r_2 r_3 r_4). \end{aligned}$$

### 6.3.3 Special polynomials

In what follows, we will illustrate the techniques of taking roots in more complicated cases by a series of examples. In this we shall often need the coefficients of  $a^r b^{n-r}$  in  $(a + b)^n$ . These are given by the binomial coefficients

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

$(a + b)^0;$

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$(a + b)^4$

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$(a + b)^2:$

$$\begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

$(a + b)^3:$

$$\begin{pmatrix} 3 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 3 \\ 1 \end{pmatrix} \quad \begin{pmatrix} 3 \\ 2 \end{pmatrix} \quad \begin{pmatrix} 3 \\ 3 \end{pmatrix}$$

$$(a + b)^4:$$

$$\binom{4}{0} \quad \binom{4}{1} \quad \binom{4}{2} \quad \binom{4}{3} \quad \binom{4}{4}$$

$$(a + b)^5:$$

$$\binom{5}{0} \quad \binom{5}{1} \quad \binom{5}{2} \quad \binom{5}{3} \quad \binom{5}{4} \quad \binom{5}{5}$$

$(a + b)^6:$

$$\binom{6}{0} \quad \binom{6}{1} \quad \binom{6}{2} \quad \binom{6}{3} \quad \binom{6}{4} \quad \binom{6}{5} \quad \binom{6}{6}$$

$(a + b)^7:$

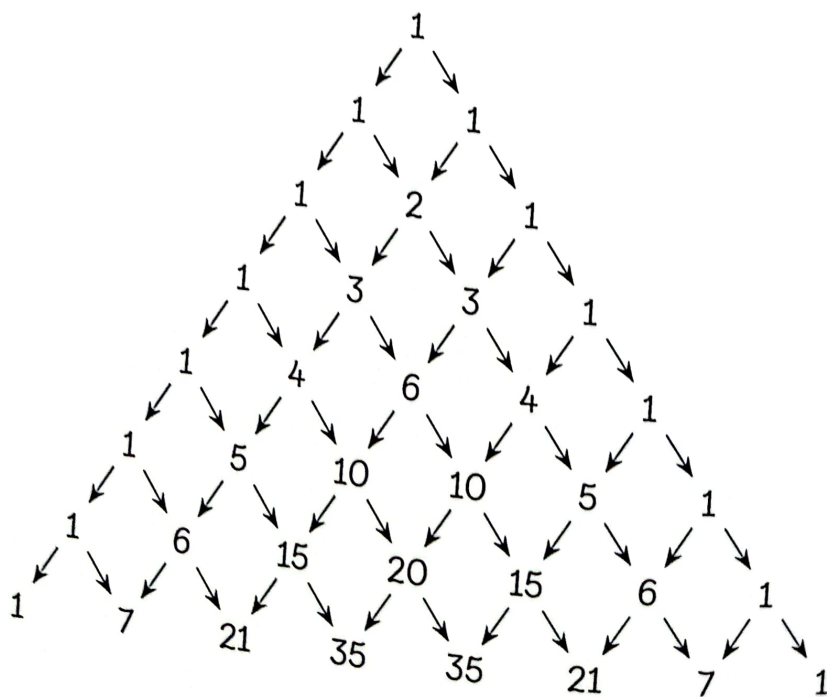
$\binom{7}{0} \quad \binom{7}{1} \quad \binom{7}{2} \quad \binom{7}{3} \quad \binom{7}{4} \quad \binom{7}{5} \quad \binom{7}{6} \quad \binom{7}{7}$

$(a + b)^8:$

(0) (1) (2) (3) (4) (5) (6) (7) (8)

Useful mnemonics for these coefficients is represented by the Pascal's triangle in which each row is obtained from the one above by adding the numbers to right and left of the position to be filled in. That is, the  $r$ -th element of  $n$ -th row is given by the sum of the two elements above it in the  $(n - 1)$ -th row:

$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r} \quad (6.30)$$



Consider the equation

$$(z + i)^7 + (z - i)^7 = 0.$$

This may be solved by the techniques just discussed as follows:

$$\left(\frac{z + i}{z - i}\right)^7 = -1 = e^{(2k+1)\pi i}$$

$$\Rightarrow \frac{z + i}{z - i} = e^{(2k+1)\pi i/7}$$

$$\Rightarrow z(1 - e^{(2k+1)\pi i/7}) = -i(1 + e^{(2k+1)\pi i/7})$$

$$\Rightarrow z = i \frac{e^{(2k+1)\pi i/7} + 1}{e^{(2k+1)\pi i/7} - 1}$$

$$= i \frac{e^{(2k+1)\pi i/14} + e^{-(2k+1)\pi i/14}}{e^{(2k+1)\pi i/14} - e^{-(2k+1)\pi i/14}}$$

$$= i \frac{2 \cos(\frac{2k+1}{14}\pi)}{2i \sin(\frac{2k+1}{14}\pi)}.$$

Thus

$$z = \cot\left(\frac{2k+1}{14}\pi\right), \quad k = 0, 1, 2, 3, 4, 5, 6.$$

The original equation can be written in another form

$$\Rightarrow z^7 - 21z^5 + 35z^3 - 7z = 0$$

$$\Rightarrow z^6 - 21z^4 + 35z^2 - 7 = 0 \text{ or } z = 0$$

$$\Rightarrow w^3 - 21w^2 + 35w - 7 = 0 \quad (w \equiv z^2)$$

Thus, using our solution for the roots of the original equation, we see the roots of

$$w^3 - 21w^2 + 35w - 7 = 0 \text{ are } w = \cot^2\left(\frac{2k+1}{14}\pi\right) (k = 0, 1, 2).$$

Sometimes the underlying equation that can be solved by these techniques is not obvious.

**EXAMPLE 6.28**

Find the roots of

$$z^3 + 7z^2 + 7z + 1 = 0.$$

Observe that the binomial coefficients for  $n = 8$  are:

$$1 \quad 8 \quad 28 \quad 56 \quad 70 \quad 56 \quad 28 \quad 8 \quad 1,$$

$$\begin{aligned} \text{so, } \frac{1}{2}[(z+1)^8 - (z-1)^8] &= 8z^7 + 56z^5 + 56z^3 + 8z \\ &= 8z[w^3 + 7w^2 + 7w + 1] \quad (w \equiv z^2). \end{aligned}$$

$$\text{Now } (z+1)^8 - (z-1)^8 = 0 \text{ when } \frac{z+1}{z-1} = e^{2k\pi i/8}, \text{ i.e. when}$$

$$z = \frac{e^{k\pi i/4} + 1}{e^{k\pi i/4} - 1} = -i \cot(k\pi/8) \quad (k = 1, 2, \dots, 7),$$

so the roots of the given equation are

$$z = -\cot^2(k\pi/8), \quad k = 1, 2, 3.$$

**Exercises**

1. In the given problems compute all the roots. Give the principal  $n$ th root in each case. Sketch the roots  $w_0, w_1, \dots, w_{n-1}$  on an appropriate circle centered at the origin.

(a)  $(8)^{1/4}$

(c)  $(3 + 3i)^{1/3}$

(b)  $(-1)^{1/5}$

(d)  $\left(\frac{1+i}{\sqrt{3}+i}\right)^{1/6}$

2. (a) Verify that  $(2 + 3i)^3 = -46 + 9i$ .

(b) Use part (a) to find the three values of  $(-46 + 9i)^{1/3}$ .

3. Find all the solutions of the equation

$$z^2 - 8z + 16 = 8i.$$

4. (a) Show that the  $n$ th roots are given by

$$(1)^{1/n} = \cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n}, \quad k = 0, 1, 2, \dots, n-1.$$

- (b) Find  $n$   $n$ th roots of unity for  $n = 3, n = 4$ , and  $n = 5$ .